scans 640x480 pixels of colour using its CMOS sensors. Its infrared projector is an IR laser that's passed through a diffraction grating that shoots IR dots — think of it as splitting light through a prism but only much cooler. The IR scanning camera is another CMOS that scans the dots shot by the IR laser and figures out how far the dot has traveled.

DOES KINECT'S LASER GIVE YOU CANCER?

Some lasers are believed to cause cancer if subjected to prolonged exposure. However, Microsoft Kinect's laser comes under the Class 1 category and is safe to use. (Read more: http://goo.gl/HfKrR).

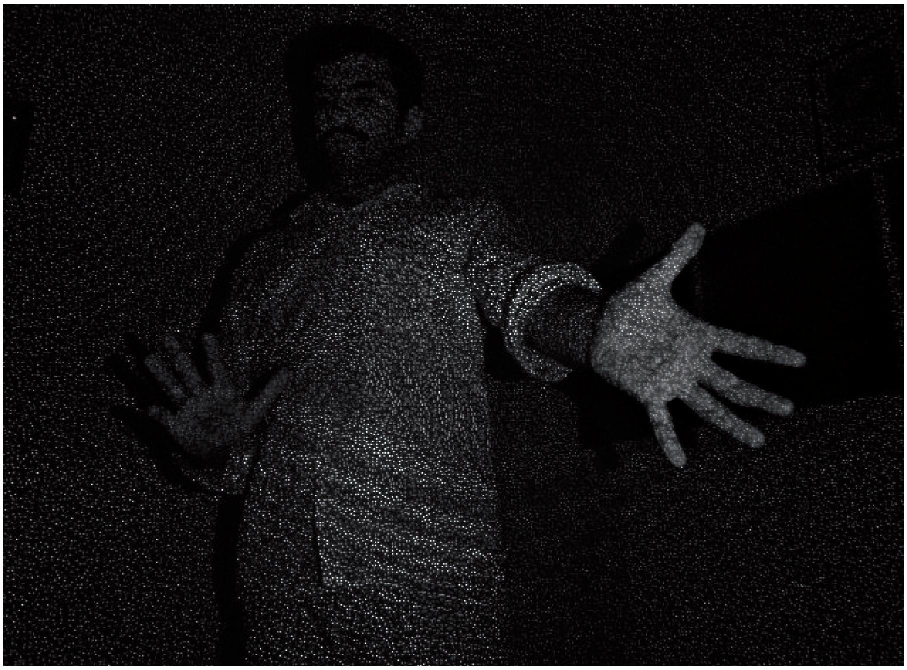
HOW DOES KINECT'S SYSTEM RECOGNISE THE USER'S POSITION?

A small history lesson before we answer that question. Computer scientists have long been trying to figure out how to detect the skeleton of an object or human from a depth map – there are decade-old papers you can read on the subject. Generally, there are two different approaches by which a software system can detect a human and calibrate the position of body parts like hands, legs and other organs.

The first method is by using a rather daft software technique whereby algorithms detect hands and other organs based on their shape and joints. This technique requires the user to stand in a particular posture called the calibration posture, during which the software measures the user's height and length of each bone for further calibration and fine-tuning. It then uses a tracking algorithm which tracks the movement of each joint. The drawbacks of this method are that it needs recalibration every time the user moves away and comes back in front of the camera. Then there's also the small matter of this technique being slow and CPU hungry. Not the best approach.

The second method estimates the position of each joint using a slightly better, more refined route. The Kinect system is fed with previous inputs - in which the user's positions are known (this is usually calibrated using other marker tracking techniques which we won't delve into in this article). The system then compares fresh input received from Kinect to the nearest known data points in its database. This search-and-retrieve technique is considered more reliable for mapping spacial gestures since having more data fed to the system means less likelihood of errors. However, this method may need plenty of storage space and an efficient searching technique as things may get slower with a larger input data bank.

With respect to Kinect, Microsoft made its system very reliable by combining both the techniques in the right proportion. Its system has simpler algorithms that usually detect just the key points from the depth map, thus reducing the errors of other mapping techniques. The advantage of its system is that it doesn't compare the depth map from Kinect directly, but detects



The Kinect doesn't see the world as the human eye. Here's how a guy waving his hand in front of the game controller paints an IR dot pattern on the Kinect's digital retina.